



HOUSE PRICE PREDICTION

PROJECT SUMMARY DOCUMENTATION

INTERN DETAILS

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ORGANIZATION

Xylofy AI
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Website: www.xylofyai.com
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PROJECT DETAILS

Project Title: House Price Prediction
Domain: Real Estate / Machine Learning
Technology: Python, Pandas, Scikit-learn, Matplotlib, Seaborn
Tools: Jupyter Notebook

SUBMISSION DETAILS

Submission Actual Date: 19-06-2026
Submission Last Date: 21-06-2026

1. PROJECT OBJECTIVE

The objective of this project is to build a machine learning model that can predict house prices based on various features such as area, number of rooms, bathrooms, parking, location-related attributes, and amenities. The project aims to help real estate stakeholders make data-driven pricing and investment decisions.

2. DATASET OVERVIEW

- Source:** Kaggle – Housing Prices Dataset
- Total Records:** 545
- Total Features:** 13 (including target variable)
- Target Variable:** Price
- Feature Types:** Numerical and Categorical

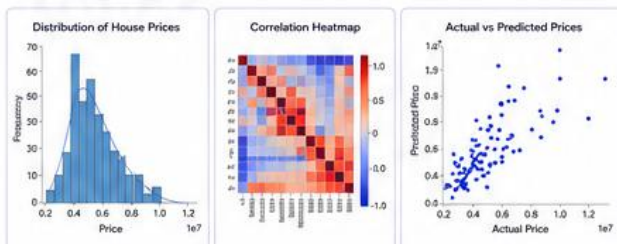
3. DATA PREPROCESSING

- Handled missing values using appropriate imputation (median/mode).
- Removed duplicate records.
- Converted categorical features into numerical format using One-Hot Encoding.
- Retained only relevant features for model training.
- Split the dataset into training (80%) and testing (20%).

4. EXPLORATORY DATA ANALYSIS

- Analyzed feature distributions and relationships.
- Identified strong correlations between features and price.
- Visualizations were used to understand patterns and outliers in the data.

KEY VISUALIZATIONS



8. CONCLUSION

This project demonstrates the end-to-end process of building a house price prediction model using machine learning. The Random Forest Regressor delivered high accuracy and can be effectively used for real-world price prediction. With more data and advanced tuning, the model's performance can be further improved.

5. MODEL BUILDING & EVALUATION

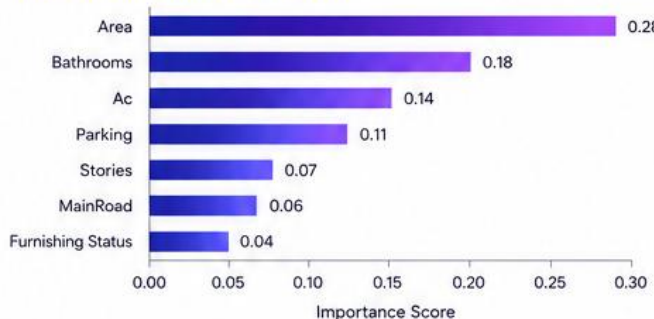
Two regression models were implemented and evaluated.

Model	MAE (↓)	RMSE (↓)	R ² Score (↑)
Linear Regression	₹ 5,23,764	₹ 6,91,245	0.72
Random Forest Regressor	₹ 2,81,736	₹ 3,91,488	0.89

Best Model: Random Forest Regressor

The Random Forest Regressor performed better than Linear Regression across all evaluation metrics, indicating higher accuracy and better generalization on unseen data.

FEATURE IMPORTANCE (TOP 7)



6. KEY FINDINGS

- Area is the most significant factor affecting house prices.
- Bathrooms, parking availability, air conditioning, and furnishing status also have a strong impact on price.
- Properties with modern amenities and larger spaces are priced considerably higher.
- The model can be used as a reliable tool for price estimation and market analysis.

7. BUSINESS RECOMMENDATION

Real estate businesses should focus on key value-adding features such as larger area, more bathrooms, parking facilities, and modern amenities while marketing properties. These factors significantly influence buyer decisions and property value.

PROJECT DELIVERABLES

- ✓ analysis.ipynb – Jupyter Notebook
- ✓ Housing.csv – Dataset Used
- ✓ summary.pdf – Project Summary
- ✓ charts/ – Visualizations (PNG)

CONTACT INFORMATION

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